
Algorithm 3: A PPN study

input: $\mathbf{x}_{\text{obs}} = \{\mathbf{x}_{\text{in}}, \mathbf{x}_{\text{out}}, \mathbf{x}_{\text{val}}\}$, models $\{\mathcal{M}_k\}_{k=1}^K$, diagnostics $\{d_k(\cdot; \mathbf{x}_{\text{val}})\}_{k=1}^K$
output: A collection of PPNs
 $S = \emptyset$;
for $k = 1, \dots, K$ **do**
 compute the empirical heldout predictive check for $\mathcal{M}_k$ ([Algorithm 1](#));
 if $\mathcal{M}_k$ passes the check **then**
 $S \leftarrow S \cup \{k\}$
for $k \in S$ **do**
 for $j \in S \setminus \{k\}$ **do**
 compute $\text{PPN}(d_k, p_k, p_j, \mathbf{x}_{\text{obs}})$ ([Algorithm 2](#));

$$y_i | \theta, \beta, \mathbf{x}_i, \mathcal{M}_B \sim N(\theta + \mathbf{x}_i^\top \beta, 1), \quad p(\theta, \beta) \propto 1, \quad (15)$$

with $\theta \in \mathbb{R}$ and $\beta, \mathbf{x}_i \in \mathbb{R}^p$.

Given observed data $(\mathbf{y}_{\text{obs}}, \mathbf{X}_{\text{obs}})$, a PPN study helps answer the questions: Do models A and B adequately capture the data? Is model A sufficient to model the data or is the more complex model B required? In detail, the study follows these steps:

1. Split the data into $(\mathbf{y}_{\text{obs}}, \mathbf{X}_{\text{obs}}) = \{(\mathbf{y}_{\text{in}}, \mathbf{X}_{\text{in}}), (\mathbf{y}_{\text{out}}, \mathbf{X}_{\text{out}}), (\mathbf{y}_{\text{val}}, \mathbf{X}_{\text{val}})\}$.
2. Choose a validation diagnostic,

$$d_{\mathcal{M}}(\mathbf{y}; \mathbf{y}_{\text{val}}) = \sum_{i=1}^n (y_i - \mathbb{E}[y_i | \mathbf{y}_{\text{val}}; \mathcal{M}])^2. \quad (16)$$

3. Calculate the posterior predictive distributions for both $\mathcal{M}_A$ and $\mathcal{M}_B$ given the in-sample data $\mathbf{y}_{\text{in}}$: $p(\mathbf{y}_{\text{rep}}^A | \mathbf{y}_{\text{in}}, \mathbf{X}_{\text{in}}; \mathcal{M}_A)$ and $p(\mathbf{y}_{\text{rep}}^B | \mathbf{y}_{\text{in}}, \mathbf{X}_{\text{in}}; \mathcal{M}_B)$.
4. Calculate heldout predictive p -values for both models,

$$P(d_A(\mathbf{y}_{\text{rep}}^A; \mathbf{y}_{\text{val}}) > d_A(\mathbf{y}_{\text{out}}; \mathbf{y}_{\text{val}}) | \mathbf{y}_{\text{in}}, \mathbf{X}_{\text{in}}; \mathcal{M}_A) \quad (17)$$

$$P(d_B(\mathbf{y}_{\text{rep}}^B; \mathbf{y}_{\text{val}}) > d_B(\mathbf{y}_{\text{out}}; \mathbf{y}_{\text{val}}) | \mathbf{y}_{\text{in}}, \mathbf{X}_{\text{in}}; \mathcal{M}_B). \quad (18)$$

5. Assuming both models pass their checks, calculate the PPN, which checks if posterior predictive data from model A can “fool” posterior predictive data from model B (under the diagnostic d_B),

$$\text{PPN}(d_B, p_A, p_B, \mathbf{y}_{\text{in}}) = D(p_A(d_B(\mathbf{y}_{\text{rep}}^A; \mathbf{y}_{\text{val}}) | \mathbf{y}_{\text{in}}) \parallel p_B(d_B(\mathbf{y}_{\text{rep}}^B; \mathbf{y}_{\text{val}}) | \mathbf{y}_{\text{in}})). \quad (19)$$

6. If the PPN passes, conclude that model A is consistent with the data and that model B is not required.

Suppose model A is true; the covariates are not involved in producing y . To demonstrate the PPN study, we generated 2,000 data points from this model (Eq. 14, $\theta = 2.5$) along with ten (meaningless) covariates. We then ran a PPN study to compare model A and model B; the results are in Figure 3.

We see that both models pass the heldout predictive check, the distributions of $d_B(\mathbf{y}_{\text{rep}}^A; \mathbf{y}_{\text{val}})$ and $d_B(\mathbf{y}_{\text{rep}}^B; \mathbf{y}_{\text{val}})$ are visually very similar, and their symmetric KL is 0.24. From this study, we would correctly conclude that model A is adequate and that the more complex model B (which still passes its check) is not needed.

In this simple situation, the PPN of Eq. 19 also has good theoretical properties. Given that model A is true, we can prove that the distributions of $d_B(\mathbf{y}_{\text{rep}}^A; \mathbf{y}_{\text{val}})$ and $d_B(\mathbf{y}_{\text{rep}}^B; \mathbf{y}_{\text{val}})$ are asymptotically equal; the correct model A can “fool” model B.

Proposition 1. *Suppose the data $(\mathbf{y}_{\text{obs}}, \mathbf{X}_{\text{obs}})$ is drawn from model A in Eq. 14; the covariates do not matter. Also assume the covariates satisfy the following condition*

$$\mathbf{x}_{\text{in},i}^\top [\mathbf{X}_{\text{obs}}^\top \mathbf{X}_{\text{obs}}]^{-1} \mathbf{x}_{\text{in},i} \rightarrow p/n \quad \text{as } n \rightarrow \infty. \quad (20)$$

Then as $n \rightarrow \infty$ and $p/n \rightarrow 0$, both $d_B(\mathbf{y}_{\text{rep}}^A; \mathbf{y}_{\text{val}})$ and $d_B(\mathbf{y}_{\text{rep}}^B; \mathbf{y}_{\text{val}})$ converge in distribution to $2\chi_n^2$ random variables.

Proof. See Appendix C of the Supplementary Material. □

The PPN of the proposition compares the posterior predictive distributions of model A and model B under a model-B diagnostic. It shows that these distributions are equal in the limit as $p/n \rightarrow 0$. As for the simulation, when the data is drawn from model A, the PPN detects that model B contains no further information. Note that the condition on the covariates in Eq. 20 may hold in a number of settings. One simple example is when the covariates are distributed as $\mathbf{x}_{\text{obs},i} \sim N(\mathbf{0}_p, \mathbf{I}_p)$, $i = 1, \dots, n$. (With conditions, some non-diagonal covariance matrices may also satisfy Eq. 20.)

This PPN study required the number of regression coefficients p to be much smaller than the sample size n . However, PPN studies are also appropriate when $p \gg n$. When $p \gg n$ in model B, the distributions of $d_B(\mathbf{y}_{\text{rep}}^A; \mathbf{y}_{\text{val}})$ and $d_B(\mathbf{y}_{\text{rep}}^B; \mathbf{y}_{\text{val}})$ will be different. This difference is due to model B overfitting the data (under the improper prior). This overfitting will be detected by a heldout predictive check. If there is alternative model that does not overfit, the PPN will detect whether the additional complexity of that model is needed.

3 Empirical studies

We demonstrate the PPN with several empirical studies.

- **Section 3.1:** We consider the infant temperament data of Stern et al. (1995), which was also analyzed by Gelman et al. (1996) to illustrate PPCs with realized discrepancies. Following the authors, we fit a multinomial mixture to the data.

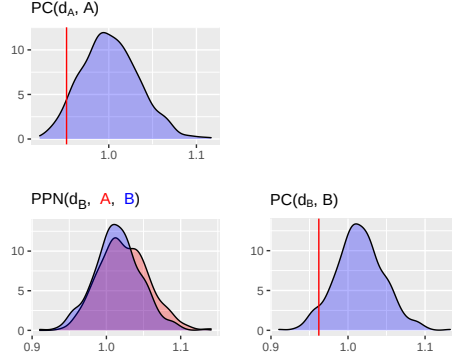


Figure 3: A PPN study of regression models. This study (correctly) suggests that model A is consistent with the data (no covariates are needed). On the diagonal are heldout predictive checks displaying $p_A(d_A(\mathbf{x}_{\text{rep}}^A; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (blue histogram) and $d_A(\mathbf{x}_{\text{out}}; \mathbf{x}_{\text{val}})$ (red line). Both models pass their checks. To the left of the diagonal is a PPN which checks if model A can fool model B. Specifically, the PPN compares $p_B(d_A(\mathbf{x}_{\text{rep}}^B; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (red histogram) and $p_A(d_A(\mathbf{x}_{\text{rep}}^A; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (blue histogram).

To choose the number of components, we use a PPN study; the result validates previous analyses of the data.

- **Section 3.2:** We consider synthetic data from a linear factor analysis model. We conduct a PPN study to choose between probabilistic PCA and two different deep generative models, fit with a variational autoencoder (VAE, [Kingma and Welling, 2014](#)) and skip-VAE ([Dieng et al., 2019](#)), respectively. The PPN study correctly suggests that PPCA is adequate to fit the data.
- **Section 3.3:** We consider synthetic data from a nonlinear factor analysis model. Here, the PPN study correctly suggests that PPCA (which assumes linearity) is *not* adequate to model the data; nonlinear deep generative models provide better fits.

3.1 Multinomial mixture model

[Stern et al. \(1995\)](#) study infant temperament data, which was also analyzed by [Gelman et al. \(1996\)](#) to illustrate PPCs with realized discrepancies. In the study, two cohorts of infants ($n = 169$, in total) were scored on the (i) degree of motor activity (scored 1-4) and (ii) crying to stimuli (scored 1-3), both at 4 months, and (iii) the degree of fear to unfamiliar stimuli at 14 months (scored 1-3). Based on these scores, it is hypothesized that infants can be clustered into two groups: inhibited and uninhibited.

To investigate the two-group hypothesis, we follow [Gelman et al. \(1996\)](#) and consider a multinomial mixture model. To choose the number of mixture components, we use a PPN study. In their analysis, [Gelman et al. \(1996\)](#) noted that “the two-class mixture